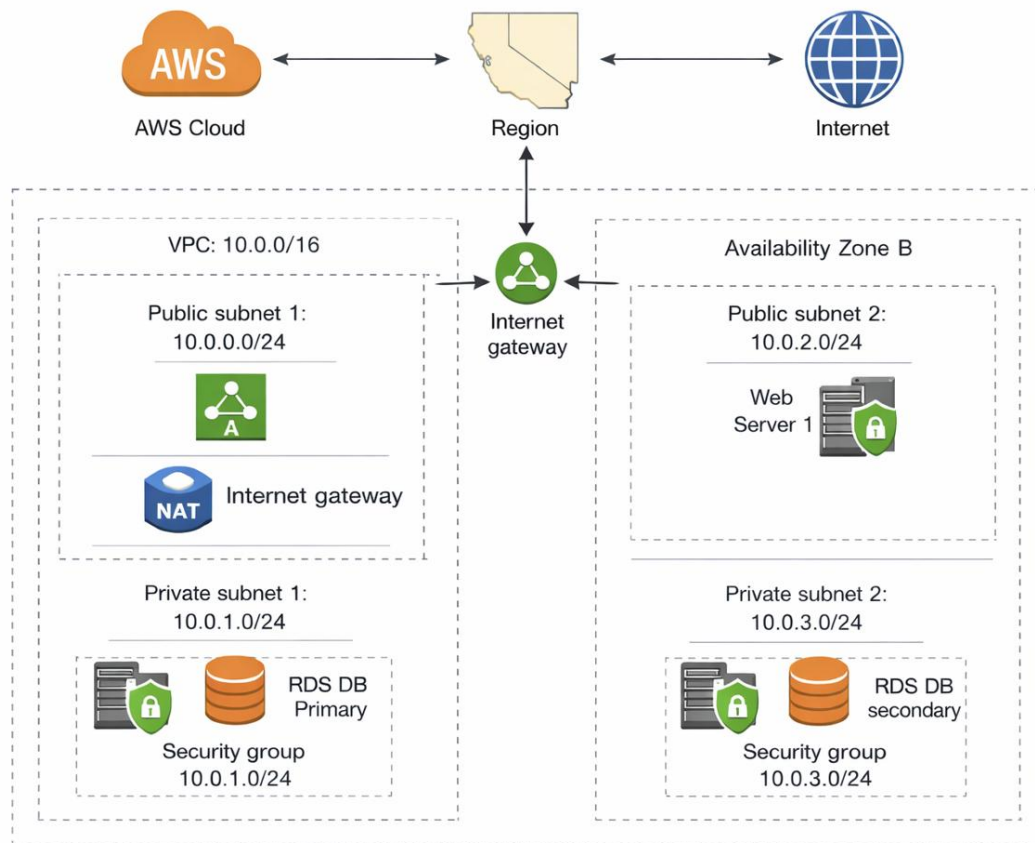
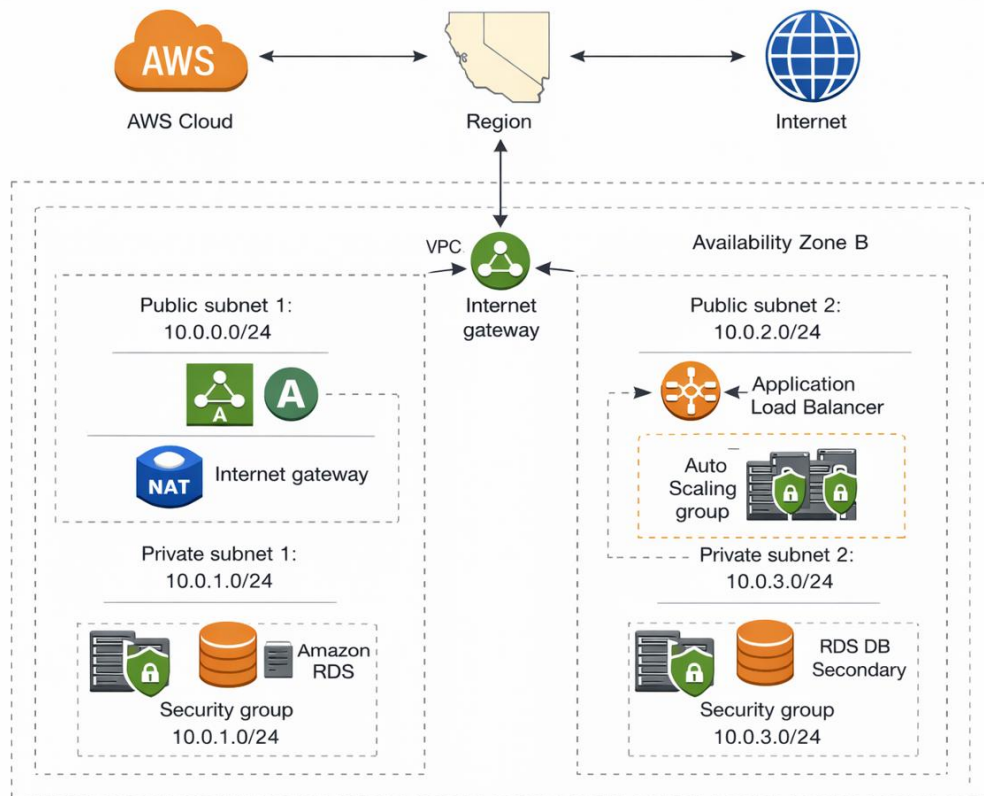


(Module 10 Lab - Scale and Load Balance Infrastructure)

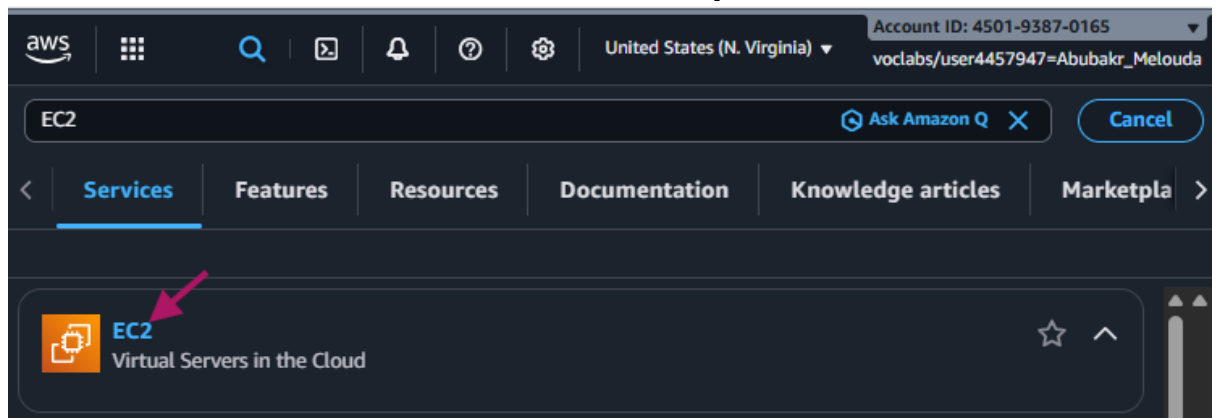
Current cloud architecture diagram

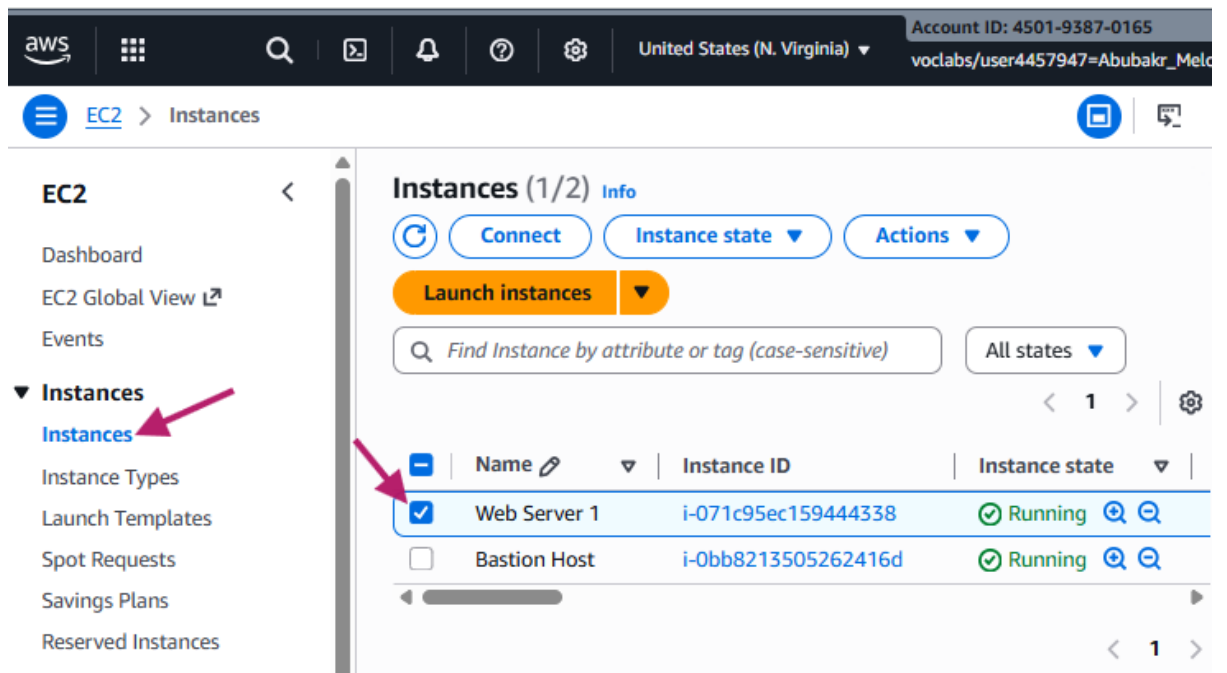


Proposed scaled and load balanced cloud architecture

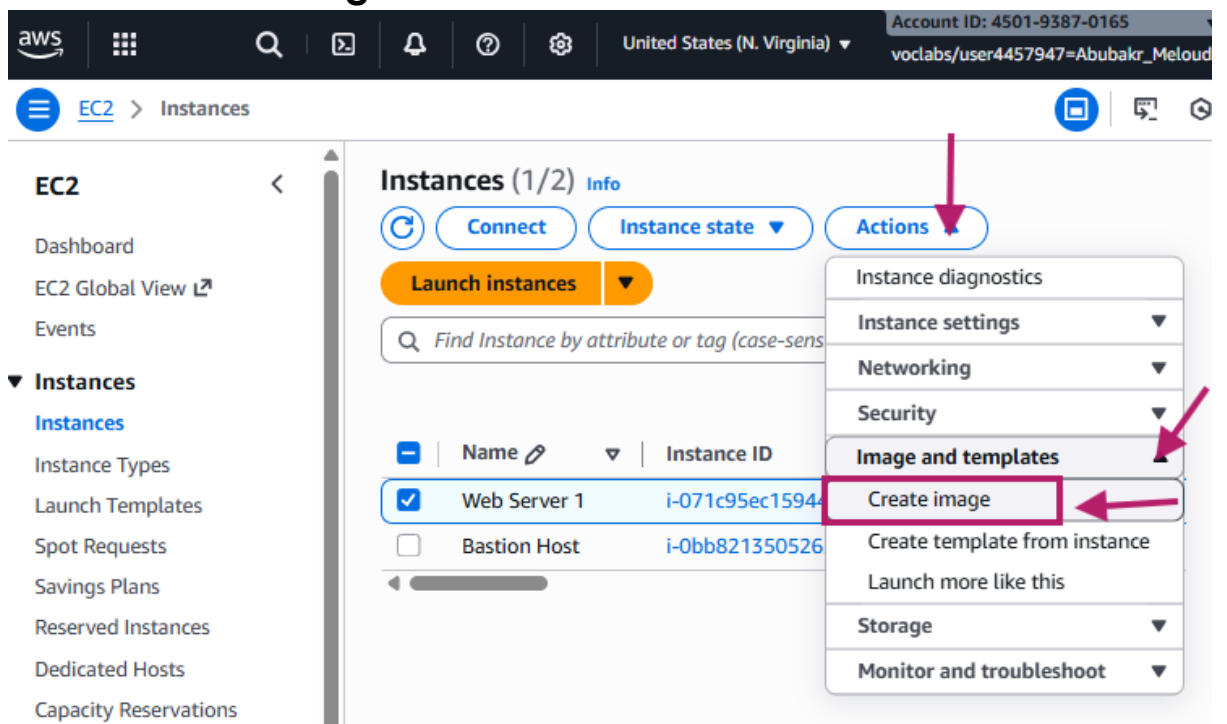


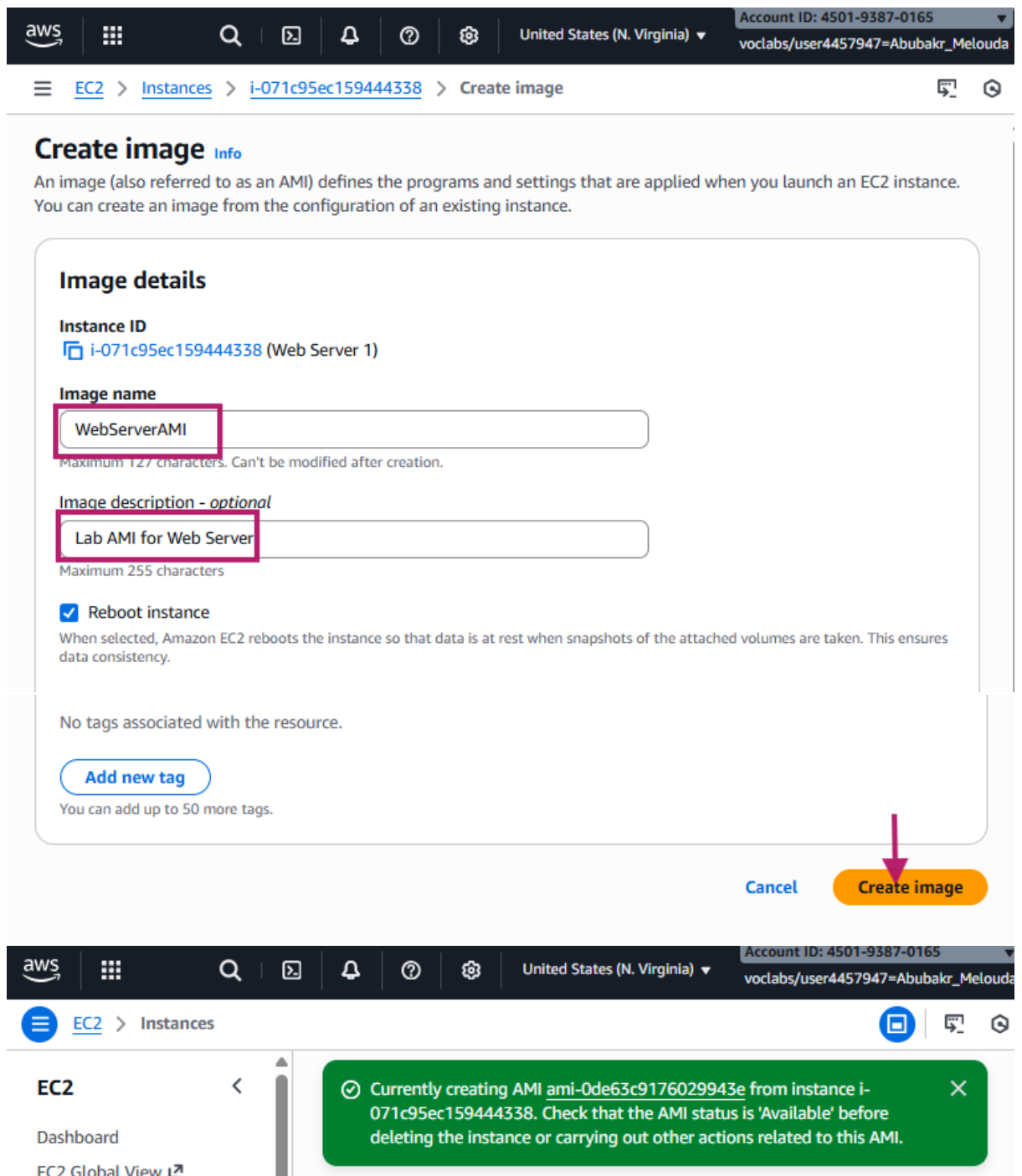
- Screenshot web Server 1 instance ready for AMI creation





- Screenshot creating AMI from Web Server 1 instance





Create image [Info](#)

An image (also referred to as an AMI) defines the programs and settings that are applied when you launch an EC2 instance. You can create an image from the configuration of an existing instance.

Image details

Instance ID
i-071c95ec159444338 (Web Server 1)

Image name
WebServerAMI
Maximum 127 characters. Can't be modified after creation.

Image description - optional
Lab AMI for Web Server
Maximum 255 characters

☒ **Reboot instance**
When selected, Amazon EC2 reboots the instance so that data is at rest when snapshots of the attached volumes are taken. This ensures data consistency.

No tags associated with the resource.

[Add new tag](#)
You can add up to 50 more tags.

[Cancel](#) [Create image](#)

EC2 <

Dashboard
EC2 Global View

Currently creating AMI ami-0de63c9176029943e from instance i-071c95ec159444338. Check that the AMI status is 'Available' before deleting the instance or carrying out other actions related to this AMI.

- Screenshot creating target group for load balancer

The screenshot displays the AWS Management Console interface for the 'Target groups' page. The top navigation bar includes the AWS logo, a search icon, a notification bell, a help icon, a settings gear, and the region 'United States (N. Virginia)'. The account ID '4501-9387-0165' and the user 'voclabs/user4457947=Abubakr_Melouda' are visible in the top right corner.

The left sidebar shows the navigation menu with the following categories:

- Instance types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager [New](#)
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store
 - Volumes
 - Snapshots
 - Lifecycle Manager
- ▼ Network & Security
 - Security Groups
 - Elastic IPs
 - Placement Groups
 - Key Pairs
 - Network Interfaces
- ▼ Load Balancing
 - Load Balancers
 - Target Groups** (highlighted with a red arrow)

The main content area is titled 'Target groups' with links for 'Info' and 'What's new?'. It features a 'Create target group' button (highlighted with a red arrow) and a search bar labeled 'Filter target groups'. Below the search bar, there is a table with columns for 'Name', 'ARN', and 'Port'. The table is currently empty, displaying the message 'No target groups' and 'You don't have any target groups in us-east-1'. A 'Create target group' button is also present below the message. At the bottom, it shows '0 target groups selected' and a prompt to 'Select a target group above.'.

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB
GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB
GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

LabGroup

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 8/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

aws

Q

United States (N. Virginia) ▼

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Target groups > Create target group

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-0de3359ccbf43a72a (Lab VPC)
10.0.0.0/16



 [Create VPC](#)

Protocol version

☒ HTTP1

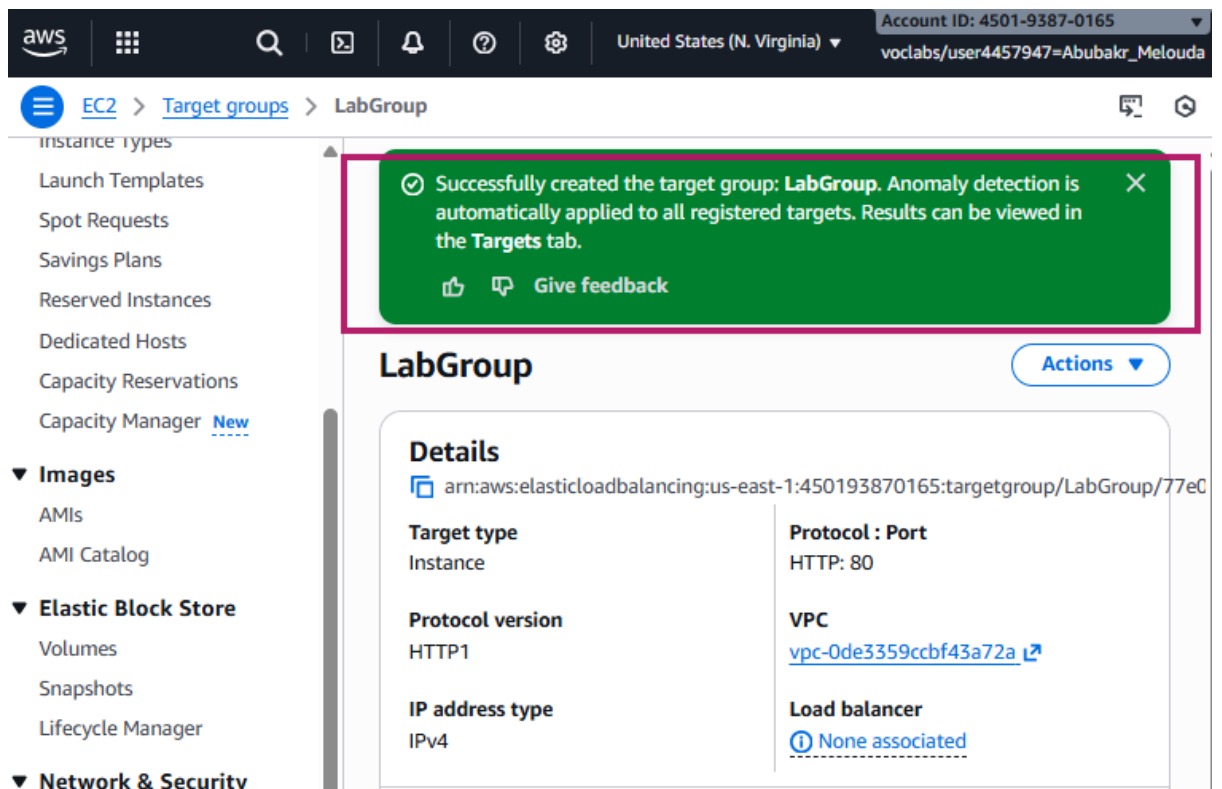
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.



- Screenshot selecting Application Load Balancer type

The screenshot displays the AWS Management Console interface. At the top, the header bar shows the AWS logo, a search icon, a help icon, a settings icon, and the region 'United States (N. Virginia)'. The account ID '4501-9387-0165' and the user 'voclabs/user4457947=Abubakr_Melouda' are also visible.

The left-hand navigation pane is expanded, showing the 'EC2' category. Under 'EC2', the 'Load balancers' link is selected. Below this, the navigation menu lists various services: Instance types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The 'Load Balancing' section is expanded, and 'Load Balancers' is highlighted with a red arrow.

The main content area is titled 'Load balancers' with a 'What's new?' link. Below the title, there is a 'Create load balancer' button, which is highlighted with a red arrow. A description states: 'Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.' Below this is a search bar labeled 'Filter load balancers'.

A table header is visible, showing columns for 'Name', 'State', 'Type', and 'Scheme'. Below the header, there is a large illustration of a robot standing next to a cloud. At the bottom of the main content area, it says '0 load balancers selected' and 'Select a load balancer above.'

aws | United States (N. Virginia) | Account ID: 4501-9387-0165 | voclabs/user4457947=Abubakr_Melo

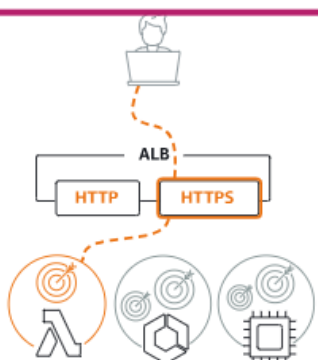
EC2 > Load balancers > Compare and select load balancer type

Compare and select load balancer type

A complete feature-by-feature comparison along with detailed highlights is also available. [Learn more](#)

Load balancer types

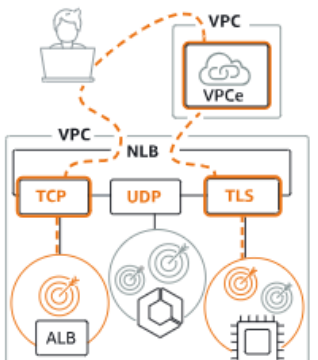
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

► Classic Load Balancer - *previous generation*

- Screenshot load balancer configuration with subnets and security group"

       United States (N. Virginia) ▼ Account ID: 4501-9387-0165 voclabs/user4457947=Abubakr_Melouda

 [EC2](#) > [Load balancers](#) > [Create Application Load Balancer](#)   

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

 **How Application Load Balancers work**

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

LabELB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme | [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the **IPv4** and **Dualstack** IP address types.

Load balancer IP address type | [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ **IPv4**
Includes only IPv4 addresses.

☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Load balancers > Create Application Load Balancer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0de3359ccbf43a72a (Lab VPC)
10.0.0.0/16

Create VPC

IP pools [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view **Pools** in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az4)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0a2641074255ea3a1
IPv4 subnet CIDR: 10.0.0.0/24

Public Subnet 1

☒ **us-east-1b (use1-az6)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-002f7c8c18c30e721
IPv4 subnet CIDR: 10.0.2.0/24

Public Subnet 2

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Load balancers > Create Application Load Balancer

Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Web Security Group

☒ **Web Security Group**
sg-0f8aa092e26d76de3 VPC: vpc-0de3359ccbf43a72a

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Load balancers > Create Application Load Balancer

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

Web Security Group
sg-0f8aa092e26d76de3 VPC: vpc-0de3359ccbf43a72a

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group Info

Choose a target group and specify routing weight or [create target group](#).

Target group

LabGroup
Target type: Instance, IPv4 | Target stickiness: Off

HTTP

Weight

1

0-999

Percent

100%

Creation workflow and status

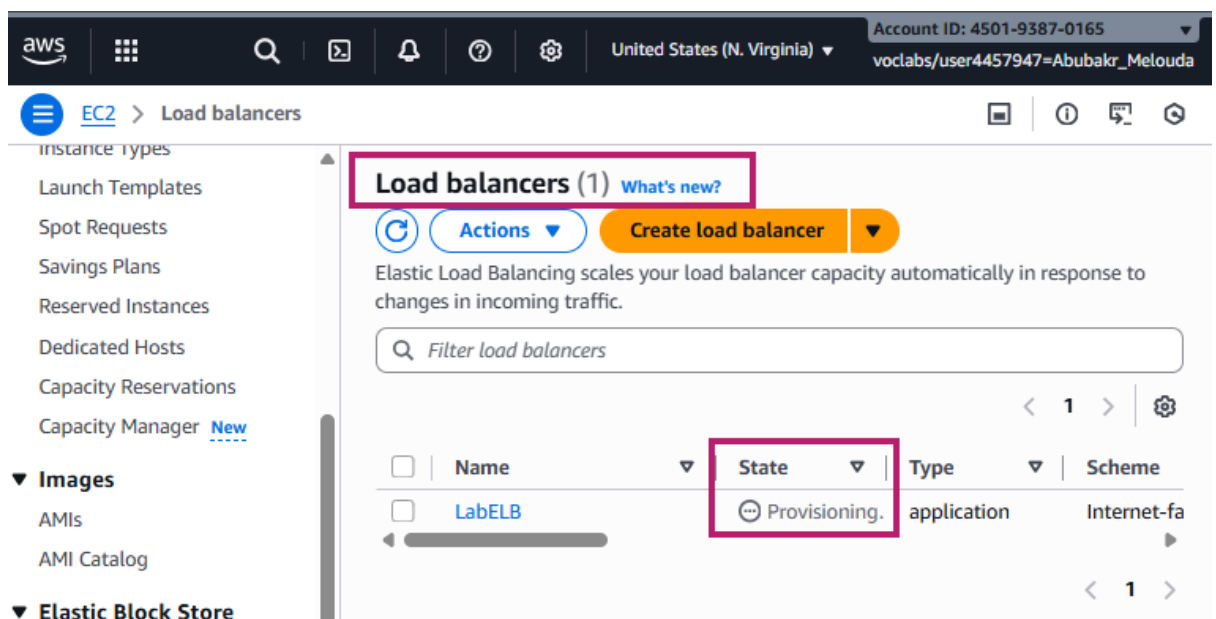
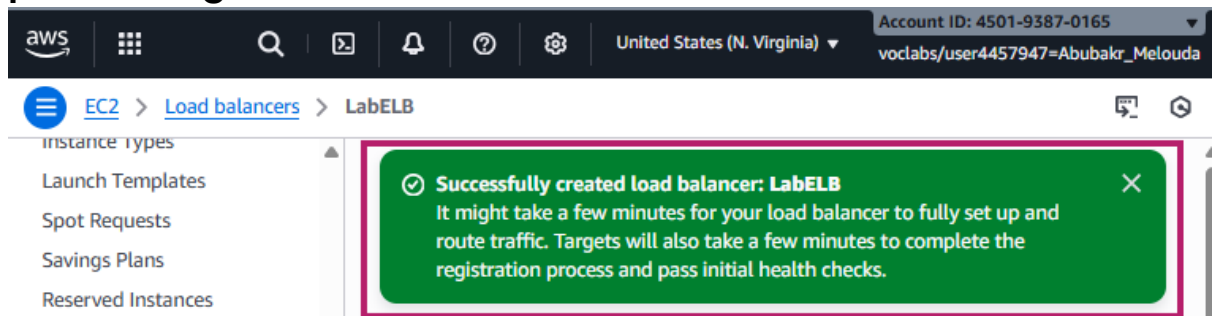
► Server-side tasks and status

After completing and submitting the above steps, all server-side tasks and their statuses become available for monitoring.

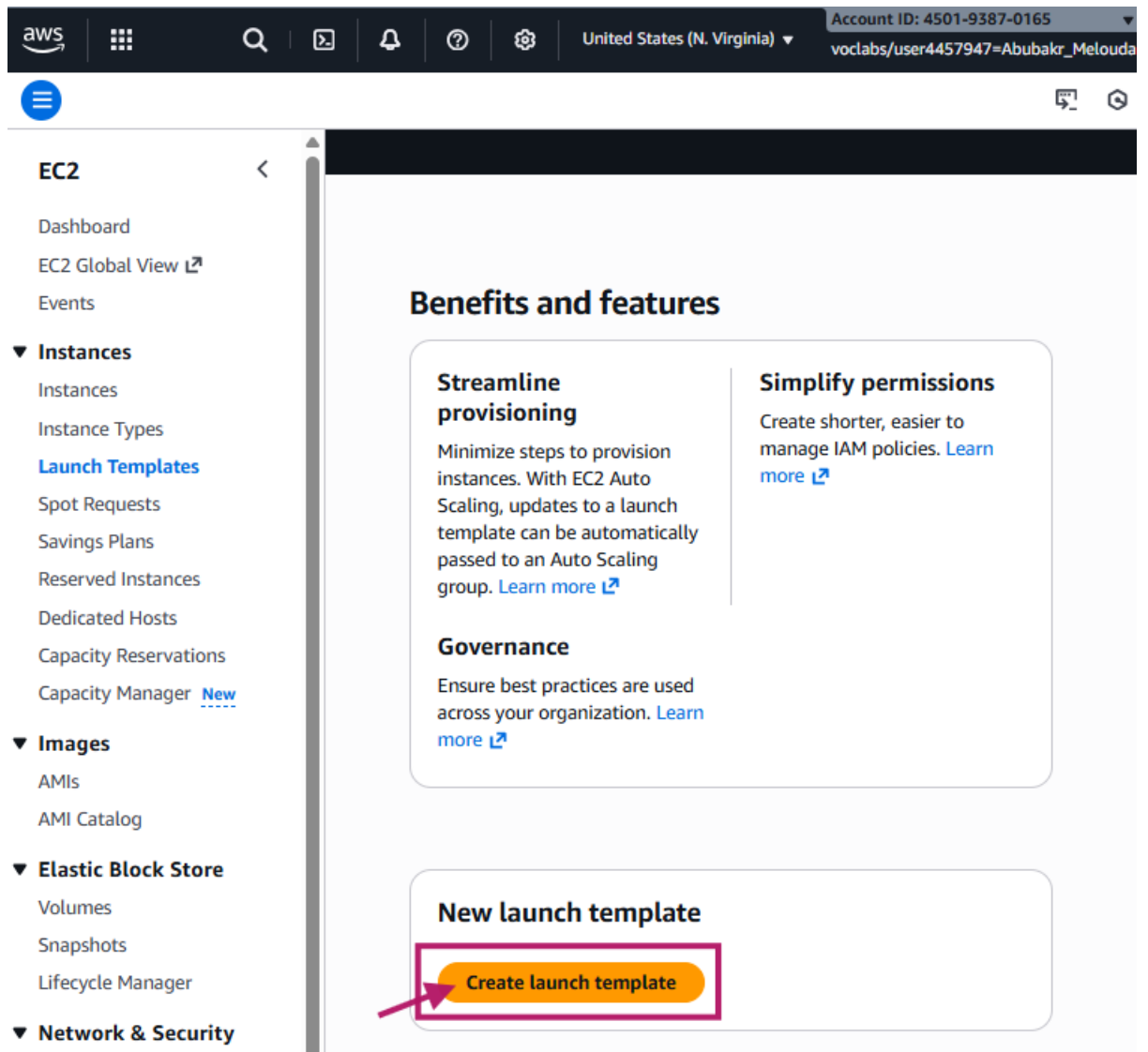
Cancel

Create load balancer

- Screenshot load balancer successfully created and provisioning



- Screenshot creating launch template with AMI and monitoring



- Screenshot launch template successfully created

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2

Launch templates

Create launch template

?

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

LabConfig

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance

[Info](#)
Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

United States (N. Virginia) ▼Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

[EC2](#) > [Launch templates](#) > Create launch template

▼ **Application and OS Images (Amazon Machine Image) - required** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents

My AMIs

Quick Start

☒ Owned by me

☐ Shared with me


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

WebServerAMI
ami-0de63c9176029943e
2026-01-08T11:59:26.000Z Virtualization: hvm ENA enabled: true Root device type: ebs Boot mode: uefi-preferred

Description
Lab AMI for Web Server

Architecture	AMI ID
x86_64	ami-0de63c9176029943e

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Launch templates > Create launch template

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

vockey

▼

↻

 Create new key pair

▼ Network settings Info

Subnet Info

Don't include in launch template

▼

↻

 Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone Info

Don't include in launch template

▼

↻

 Enable additional zones

Not applicable for EC2 Auto Scaling

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups Info

Select security groups

▼

Web Security Group sg-0f8aa092e26d76de3

VPC: vpc-0de3359ccbf43a72a

×

↻

 Compare security group rules

► Advanced network configuration

aws

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Launch templates > Create launch template

▼ Advanced details Info

IAM instance profile Info

Don't include in launch template ▼

Create new IAM profile

Hostname type Info

Don't include in launch template ▼

DNS Hostname Info

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery Info

Don't include in launch template ▼

Shutdown behavior Info

Don't include in launch template ▼

Not applicable for EC2 Auto Scaling

Stop - Hibernate behavior Info

Don't include in launch template ▼

Not applicable for Amazon EC2 Auto Scaling.

Termination protection Info

Don't include in launch template ▼

Stop protection Info

Don't include in launch template ▼

Detailed CloudWatch monitoring Info

Enable ▼

Additional charges apply

Credit specification Info

Don't include in launch template ▼

aws

United States (N. Virginia)

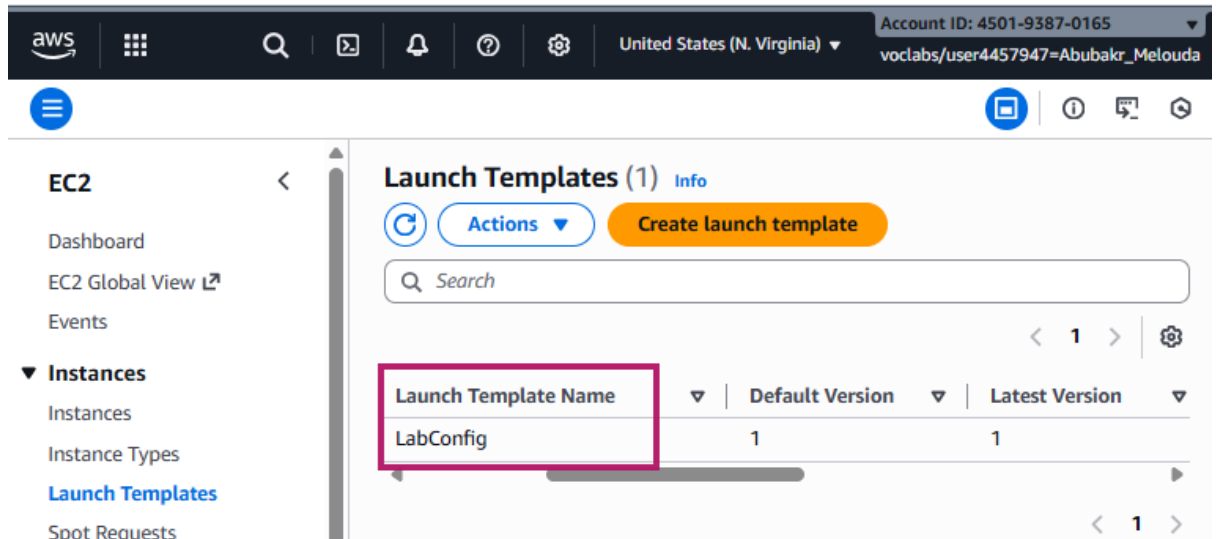
Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Launch templates > Create launch template

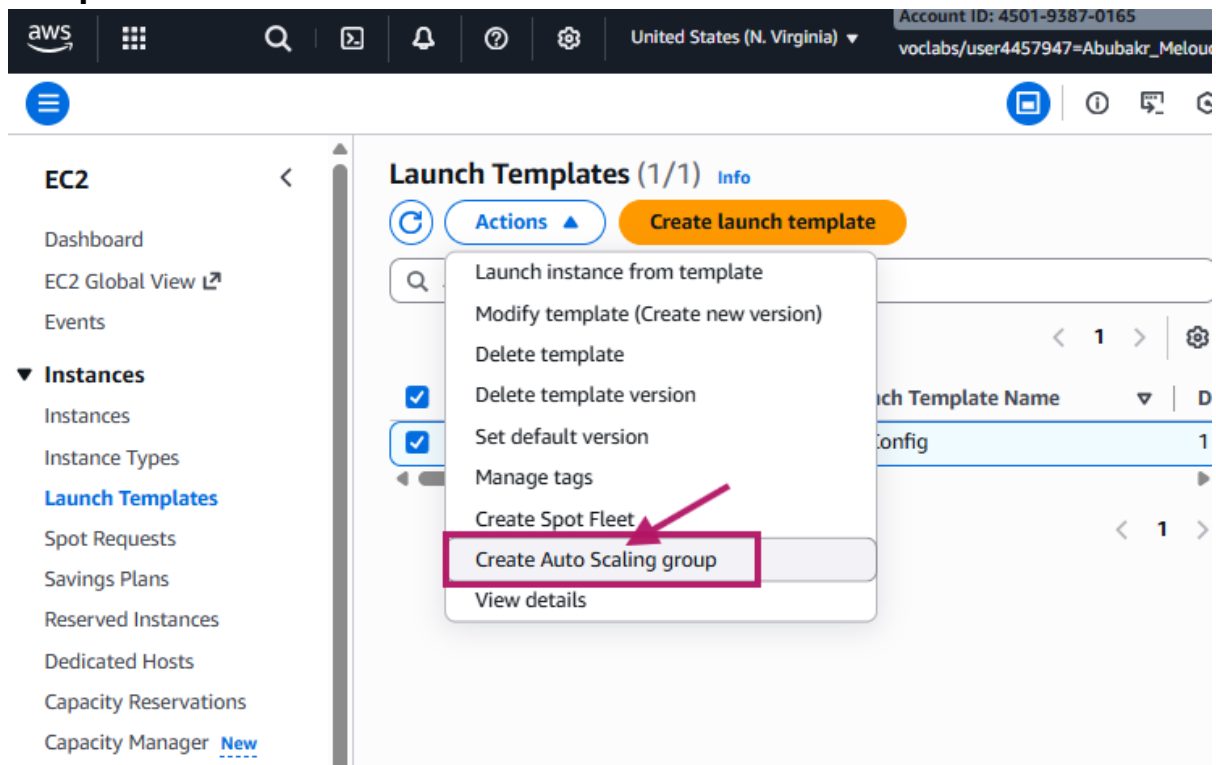
✓ Success

Successfully created [LabConfig\(lt-Od230a08d42aad3d3\)](#).

► Actions log



- Screenshot starting Auto Scaling group creation with launch template



aws [Search] [Alt+S] United States (N. Virginia) Account ID: 4501-9387-0165 voclabs/user4457947=Abubakr_Melouda

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template or configuration

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Choose launch template or configuration Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name
Enter a name to identify the group.

Lab Auto Scaling Group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info [Switch to launch configuration](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

LabConfig

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description -	Launch template LabConfig lt-0d230a08d42aad3d3	Instance type t2.micro
AMI ID ami-0de63c9176029943e	Security groups -	Request Spot Instances No
Key pair name vockey	Security group IDs sg-0f8aa092e26d76de3	

Additional details

Storage (volumes) -	Date created Thu Jan 08 2026 12:20:46 GMT+0000 (Greenwich Mean Time)
-------------------------------	---

[Cancel](#) [Next](#)

- Screenshot selecting private subnets for Auto Scaling group instances

Step 2: Choose instance launch options

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements

Launch template: LabConfig
Version: Default
Description: It-0d230a08d42aad3d3
Instance type: t2.micro

Network

VPC: vpc-0de3359ccbf43a72a (Lab VPC)

Availability Zones and subnets:

- use1-az4 (us-east-1a) | subnet-04da985c434c44b82 (Private Subnet 1)
- use1-az6 (us-east-1b) | subnet-04a71348a1b2f5462 (Private Subnet 2)

Availability Zone distribution - new

Balanced best effort (selected): If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only: If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Buttons: Cancel, Skip to review, Previous, Next

- Screenshot attaching auto scaling group to load balancer target group

Step 3 - optional
Integrate with other services

Integrate with other services - optional [Info](#)
Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)
Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

- ☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.
- ☒ **Attach to an existing load balancer**
Choose from your existing load balancers.
- ☐ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to an existing load balancer

Select the load balancers to attach

- ☒ **Choose from your load balancer target groups**
This option allows you to attach Application, Network, or Gateway Load Balancers.
- ☐ Choose from Classic Load Balancers

Existing load balancer target groups
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

- LabGroup | HTTP** [X](#)
Application Load Balancer: LabELB

VPC Lattice integration options [Info](#)
To improve networking capabilities and scalability, integrate your Auto Scaling group with VPC Lattice. VPC Lattice facilitates communications between AWS services and helps you connect and manage your applications across compute services in AWS.

Select VPC Lattice service to attach

- ☒ **No VPC Lattice service**
VPC Lattice will not manage your Auto Scaling group's network access and connectivity with other services.
- ☐ Attach to VPC Lattice service
Incoming requests associated with specified VPC Lattice target groups will be routed to your Auto Scaling group.

[Create new VPC Lattice service](#)

Health check grace period [Info](#)
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

- Screenshot configuring auto scaling policy for 60% CPU utilization

Step 4 - optional
Configure group size and scaling

Configure group size and scaling - optional [Info](#)
 Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)
 Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type
 Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity
 Specify your group size.

2

Scaling [Info](#)
 You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
 Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity **Max desired capacity**

2 6

Equal or less than desired capacity Equal or greater than desired capacity

Automatic scaling - optional
 Choose whether to use a target tracking policy [Info](#)
 You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
 Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ **Target tracking scaling policy**
 Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

LabScalingPolicy

Metric type [Info](#)
 Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization ▼

Target value

60

Instance warmup [Info](#)
 300 seconds

☐ Disable scale in to create only a scale-out policy

Default instance warmup [Info](#)
 The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

Cancel Skip to review Previous **Next**

- Screenshot adding tags to auto scaling group instances

aws | United States (N. Virginia) | Account ID: 4501-9387-0165 | voclabs/user4457947=Abubakr_Melouda

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template or configuration

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: **Add tags**

Step 7: Review

Add tags - optional Info

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

Tip: You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

Tags (1)

Key
Name

Value - optional
Lab Instance

Tag new instances
☒

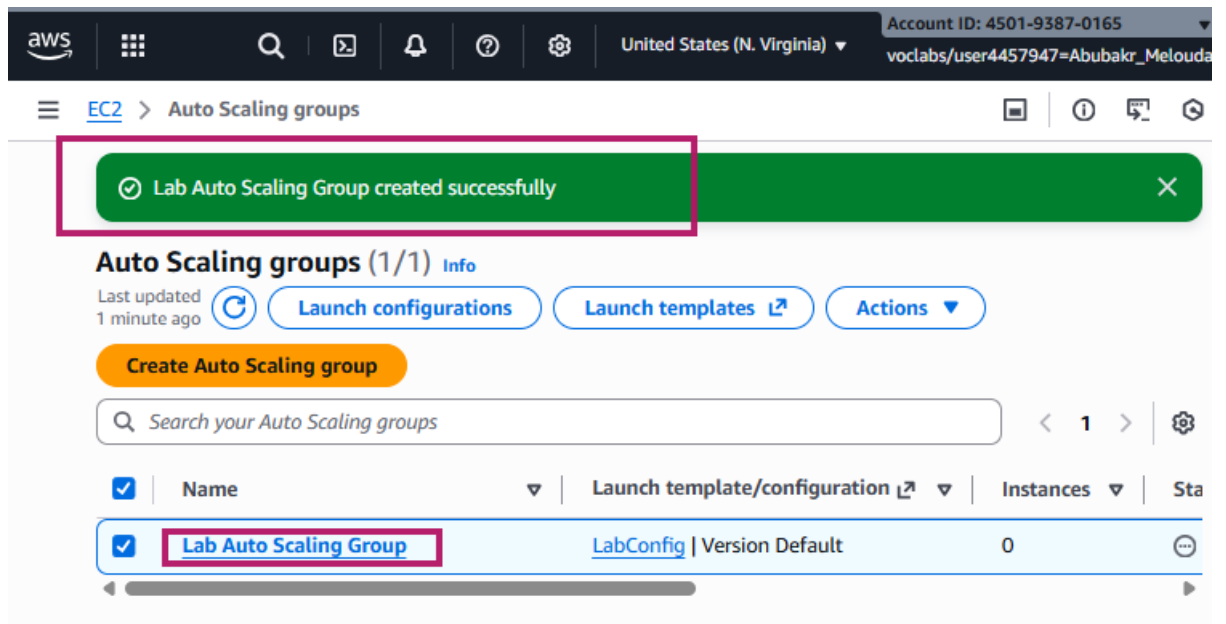
[Remove](#)

[Add tag](#)

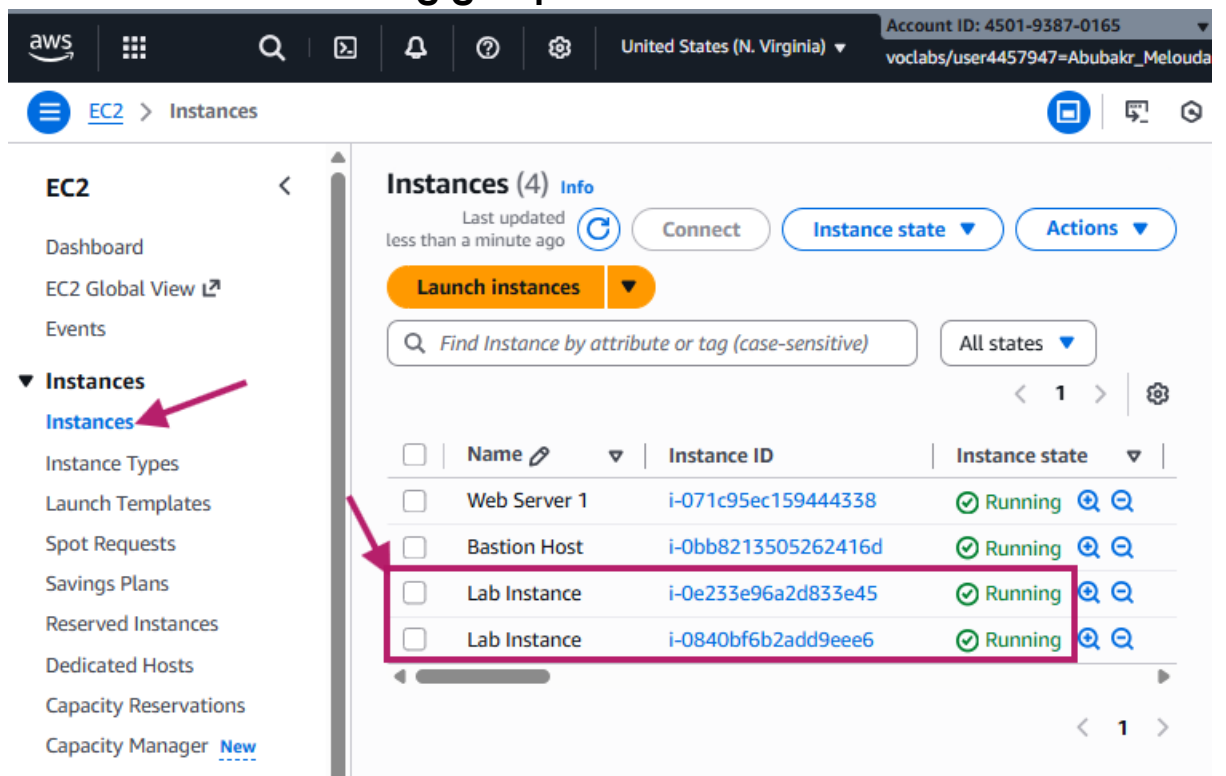
49 remaining

[Cancel](#) [Previous](#) [Next](#)

- Screenshot auto scaling group successfully created



- Screenshot auto scaling group launched 2 instances



- Screenshot load balancer target group showing healthy instances

The screenshot displays the AWS Management Console interface for managing Target Groups. The left-hand navigation pane shows the 'Target Groups' link under the 'Load Balancing' section, highlighted with a red arrow. The main content area shows the 'Target groups (1/1)' page. A table lists the target groups, with 'LabGroup' selected and highlighted by a red arrow. Below this, the 'Target group: LabGroup' details are shown, with the 'Targets' tab selected. This tab displays 'Registered targets (2)', both of which are 'Healthy', as indicated by green checkmarks in the 'Health status' column, also highlighted by red arrows. The console header shows the user is logged in as 'voclabs/user4457947=Abubakr_Melouda' in the 'United States (N. Virginia)' region.

Name	ARN	Port	Protocol	Target type
LabGroup	arn:aws:elasticloadbalancing...	80	HTTP	Instance

Instance ID	Name	Port	Zone	Health status
i-0840bf6b2add9eee6	Lab Instance	80	us-east-1...	Healthy
i-0e233e96a2d833e45	Lab Instance	80	us-east-1...	Healthy

- Screenshot web application accessible through load balancer - proof of load balancing

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

EC2 > Load balancers

Instance types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager [New](#)

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

▼ Auto Scaling
Auto Scaling Groups

Settings

Load balancers (1/1) [What's new?](#)

[Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☒	Name	State	Type	Scheme
☒	LabELB	Active	application	Internet-facing

Load balancer: LabELB

Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K
VPC vpc-0de3359ccbf43a72a	Load balancer IP address type IPv4
Availability Zones subnet-002f7c8c18c30e721 us-east-1b (use1-az6) subnet-0a2641074255ea3a1 us-east-1a (use1-az4)	Date created January 8, 2026, 12:12 (UTC+00:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:450193870165:loadbalancer/application/LabELB/969df7cda88082f5	DNS name LabELB-556186427.us-east-1.elb.amazonaws.com (A Record)

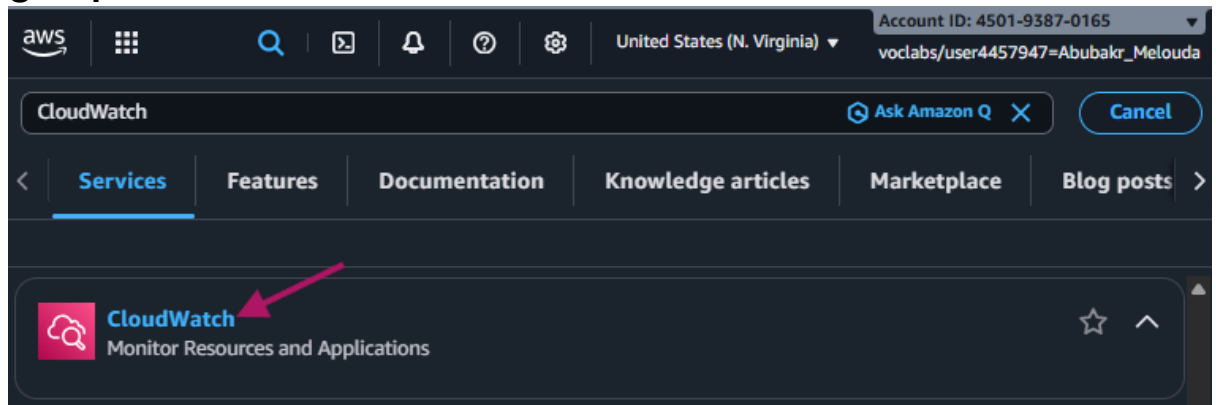
✔ DNS name copied

Load Test RDS

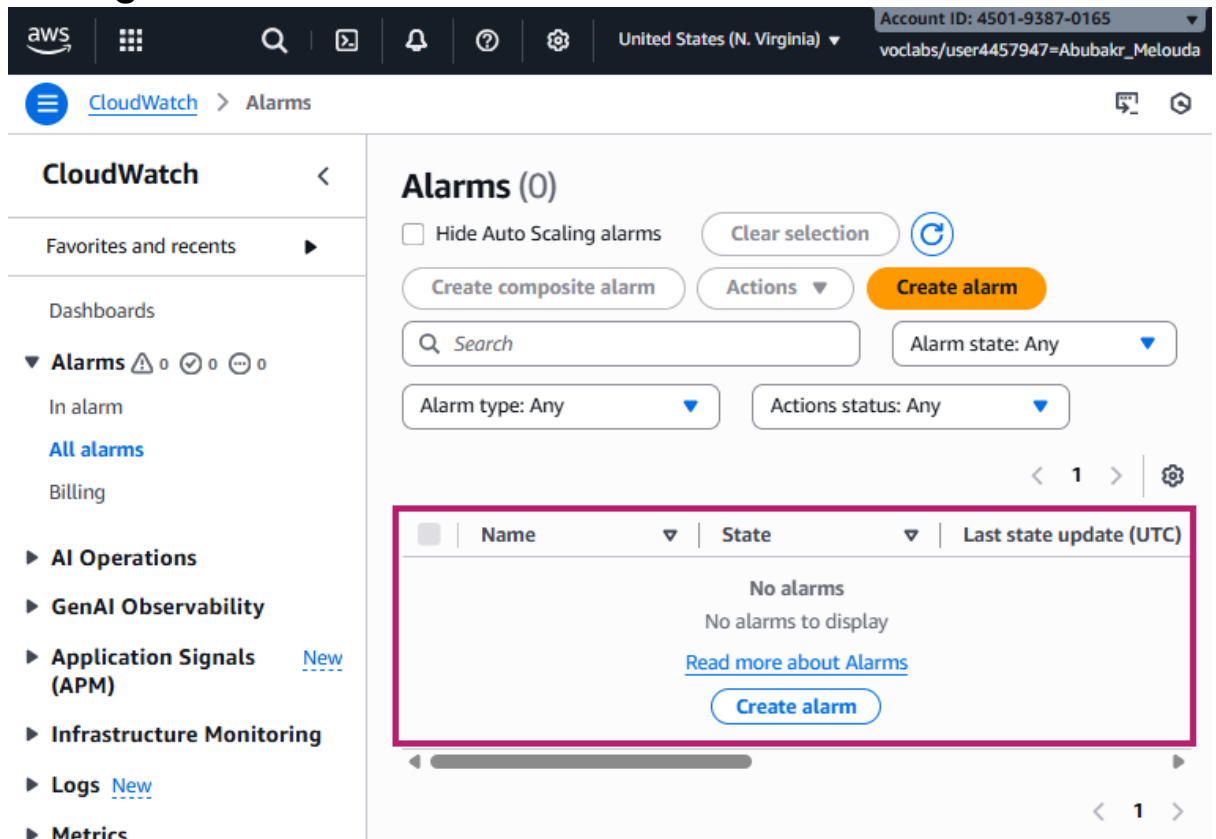
Under High CPU Load! (auto refresh in 5 seconds)

Current CPU Load: 100%

- Screenshot CloudWatch alarms created by Auto Scaling group



Screenshot troubleshooting alarms not created by auto scaling



To resolve this issue, I have manually created the alarm in CloudWatch console:

The screenshot shows the AWS CloudWatch Alarms console. The top navigation bar includes the AWS logo, account ID (4501-9387-0165), and region (United States (N. Virginia)). The left sidebar shows the CloudWatch menu with options like Favorites and recents, Dashboards, Alarms (0), In alarm, All alarms, Billing, AI Operations, GenAI Observability, Application Signals (APM), and Infrastructure Monitoring. The main content area is titled 'Alarms (0)' and features a 'Create alarm' button highlighted with a red arrow. Other buttons include 'Create composite alarm', 'Clear selection', and 'Actions'. A search bar and filters for 'Alarm type' and 'Actions status' are also present. The bottom section shows a table with columns for Name, State, and Last state update (UTC), currently displaying 'No alarms'.

The screenshot shows the 'Specify metric and conditions' step in the AWS CloudWatch 'Create alarm' wizard. The left sidebar shows the progress: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add alarm details), and Step 4 (Preview and create). The main content area is titled 'Specify metric and conditions' and features a 'Metric' section with a 'Graph' preview. A 'Select metric' button is highlighted with a red arrow. The bottom right corner has 'Cancel' and 'Next' buttons.

aws Account ID: 4501-9387-0165 voclabs/user4457947=Abubakr_Melou

United States (N. Virginia)

Select metric

Untitled graph 

3h 1d 1w  UTC timezone  Line 

1
0.5
0

Your CloudWatch graph is empty.
Select some metrics to appear here.

10:30 11:00 11:30 12:00 12:30 13:00

=

Add math  Add query 

Browse (892) | Multi source query | Graphed metrics | Options | Source

☐ Alarm recommendations   Graph with SQL  Graph search

 Search for any metric, dimension, resource id or account id

ApplicationELB	72	Bedrock/DataAutomation	5
Billing	2	EBS	33
EC2	133	Events	8

The screenshot shows the AWS CloudWatch 'Select metric' dialog box. The dialog has a title bar with 'Select metric' and a close button. Below the title bar, there is a section for 'Untitled graph' with a pencil icon, time range selectors (3h, 1d, 1w), a UTC timezone dropdown, and a Line chart type dropdown. A refresh button is also present. The main area is a graph that is currently empty, with the text 'Your CloudWatch graph is empty. Select some metrics to appear here.' Below the graph, there are buttons for 'Add math' and 'Add query'. At the bottom, there is a search bar with the placeholder text 'Search for any metric, dimension, resource id or account id'. Below the search bar, there is a list of metric categories with their respective counts: 'By Auto Scaling Group' (23), 'By Image (AMI) Id' (12), 'Per-Instance Metrics' (74), 'Aggregated by Instance Type' (12), and 'Across All Instances' (12). A red arrow points to the 'By Auto Scaling Group' option.

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

United States (N. Virginia)

Select metric

Untitled graph

3h 1d 1w

UTC timezone

Line

1
0.5
0

Your CloudWatch graph is empty.
Select some metrics to appear here.

10:30 11:00 11:30 12:00 12:30 13:00

=

Add math Add query

Browse (133) Multi source query Graphed metrics Options Source

▼ EC2 Alarm recommendations

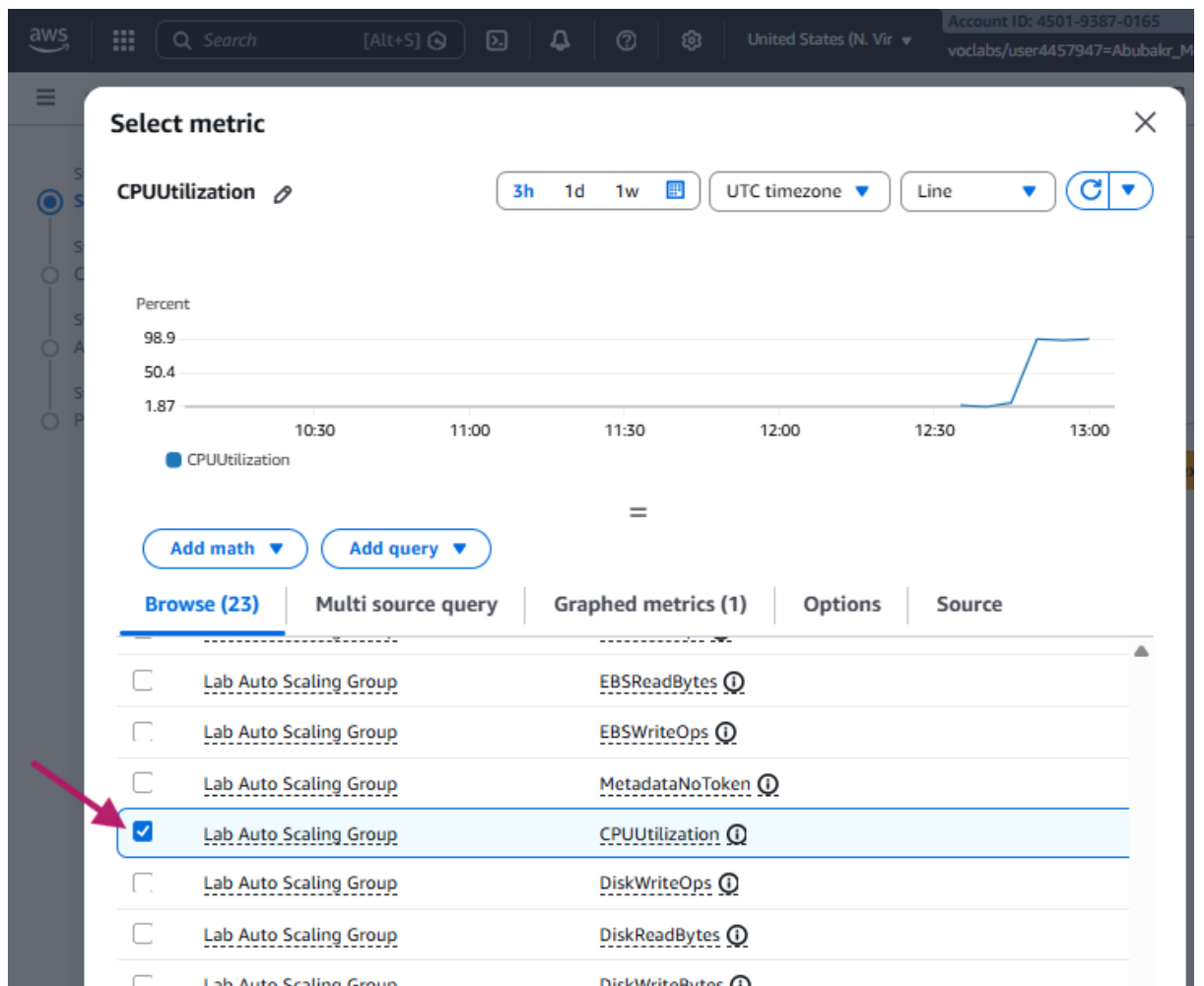
Graph with SQL Graph search

Search for any metric, dimension, resource id or account id

By Auto Scaling Group 23 By Image (AMI) Id 12

Per-Instance Metrics 74 Aggregated by Instance Type 12

Across All Instances 12



The screenshot shows the AWS CloudWatch 'Select metric' dialog. The 'CPUUtilization' metric is selected for the 'm1' instance type. The 'Add query' button is highlighted, and the 'EC2' namespace is selected. The 'Average CPU utilization across the entire fleet' query is chosen from the list of options. The 'Select metric' button is highlighted at the bottom right.

Select metric

CPUUtilization

3h 1d 1w UTC timezone Line

Percent

98.9
50.4
1.87

10:30 11:00 11:30 12:00 12:30 13:00

Add math **Add query**

Browse (23)

- ☐ Lab Auto Scaling
- ☐ Lab Auto Scaling
- ☐ Lab Auto Scaling
- ☒ Lab Auto Scaling

- Application LB
- AWS API usage examples
- EBS
- EC2**
- EventBridge
- Lambda
- Logs

metrics (1) **Options** **Source**

- ReadBytes
- CPU utilization of EC2 instances sorted by highest
- Average CPU utilization across the entire fleet**
- Top 10 instances by highest CPU utilization
- Average Active CPU for an application (CloudWatch Agent)

☒ m1 CPUUtilization EC2 • CPUUtilization • AutoScalingGroup! Avg

Cancel **Select metric**

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 4501-9387-0165
voclabs/user4457947=Abubakr_Melouda

CloudWatch > Alarms > Create alarm

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add alarm details

Step 4

Preview and create

Specify metric and conditions

Metric

Edit

Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

98.9

60

50.4

1.87

10:15 13:15

■ CPUUtilization

Namespace
AWS/EC2

Metric name
CPUUtilization

AutoScalingGroupName
Lab Auto Scaling Group

Statistic
Average

Period
5 minutes

Conditions

Threshold type

☒ Static
Use a value as a threshold

☐ Anomaly detection
Use a band as a threshold

Whenever CPUUtilization is...
Define the alarm condition.

☐ Greater
> threshold

☒ Greater/Equal
≥ threshold

☐ Lower/Equal
≤ threshold

☐ Lower
< threshold

than...
Define the threshold value.

60

Must be a number.

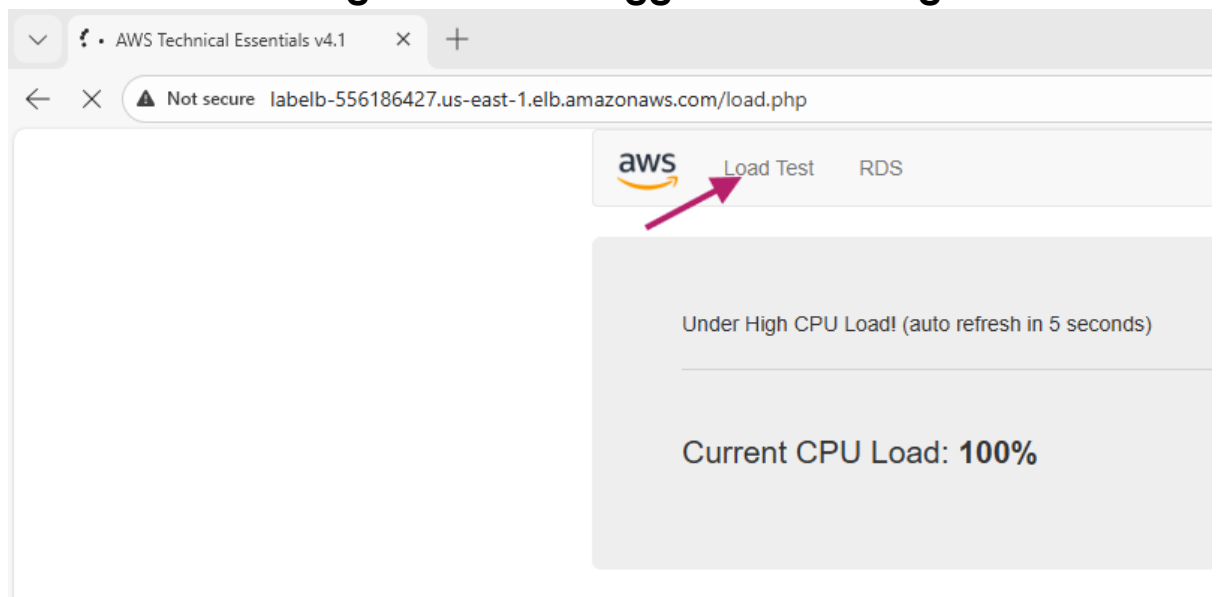
► Additional configuration

► Additional configuration

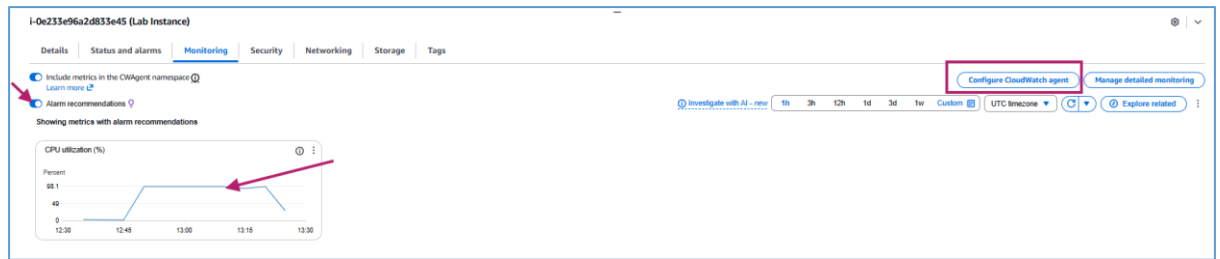
Cancel Next

The screenshot shows the AWS CloudWatch 'Create alarm' console. The 'Configure actions' step is active. Under 'Alarm state trigger', the 'In alarm' option is selected. Under 'Send a notification to the following SNS topic', the 'Create new topic' option is selected. A new topic named 'Lab-ASG-Alarm' is being created. An email endpoint 'abubakr.melouda@cloudwatch.com' is added to receive notifications. The 'Create topic' button is visible at the bottom.

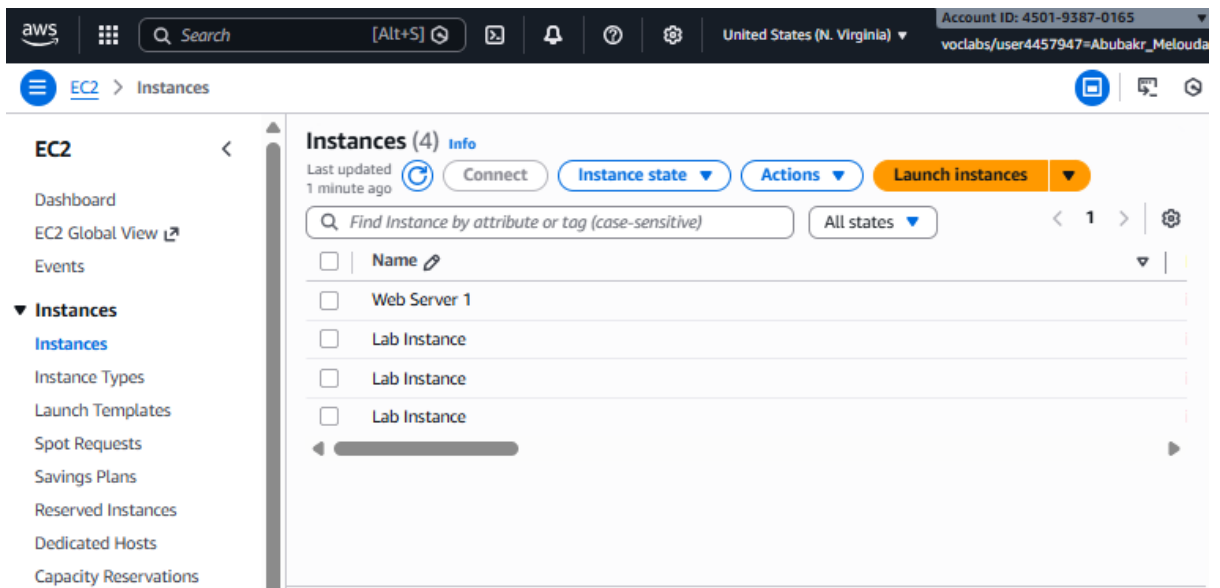
- Screenshot initiating load test to trigger auto scaling



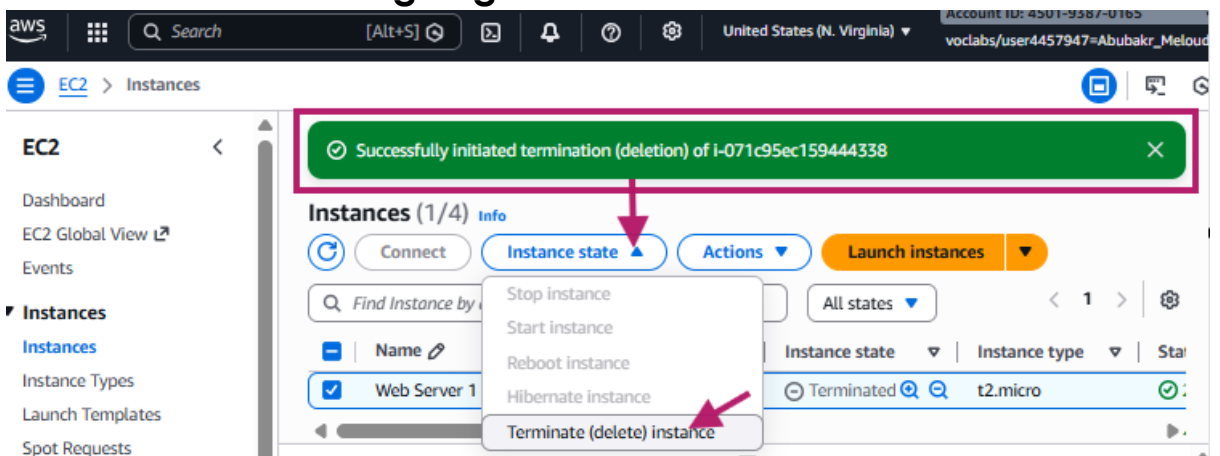
- Screenshot CloudWatch alarm triggered by high CPU load



- Screenshot auto scaling successfully launched additional instance



- Screenshot terminating original Web Server 1 instance



Lab summary:

Created auto-scaling infrastructure with load balancer. Built AMI, configured Application Load Balancer, set up Auto Scaling group with scaling policies, and tested automatic scaling under load.

Key Features:

- **AMI creation for consistent deployments**
- **Application Load Balancer with target groups**
- **Auto Scaling group (2-6 instances)**
- **Target tracking scaling policy (60% CPU)**
- **CloudWatch alarms for automatic scaling**
- **Load testing and validation**